Nathan Walker, BS Electrical Engineering

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Executive Summary

Graduate-trained Electrical Engineer with experience in digital signal processing, mathematics, PCB design, and FPGA development. Recognized as the ECE Department Outstanding Senior and consistent Dean's List student, with proven leadership through international engineering competitions. Master's research and coursework emphasize DSP hardware development and FPGA-based system design, including interactive projects using finite state machines, VGA display interfaces, and real-time user input. Known for strong communication and hands-on problem solving, with experience coordinating teams and working directly with complex technical systems.

Work Experience

Graduate Researcher May 2024 – Present

Thesis: *Design and Implementation of Adaptive Active Snore Reduction Using Near-Field Array Processing*

- Developed real-time embedded signal processing algorithms on Linux, implementing adaptive filtering and array processing for noise cancellation.
- Repaired and debugged custom electronics designed by others with no circuit documentation, identifying faults and restoring full functionality.
- Used MATLAB and C to analyze data, supporting debugging and performance optimization.
- Created custom PCB enclosures in Solid Edge to support mechanical stability and system function
- Effectively communicated complex technical problems and solutions to advising professors.
- Ensured data integrity during collection and analysis of automated experimental measurements.

ASPIRE Research Assistant May 2023 – April 2024

- Selected for the ASPIRE research program based on academic performance and faculty recommendation.
- Designed and executed PCB and component testing protocols, including soldering, signal verification, and measurement with lab instruments, ensuring accurate experimental results.
- Developed hardware solutions for research projects under size, cost, and performance constraints.
- Assembled complex PCBs and coordinated multi-component research projects from design to execution, demonstrating hands-on troubleshooting and project management.

Engineering Projects

International Future Energy Challenge – Team Lead July 2023 – July 2024

- Led a team of 3 undergraduates, taking primary responsibility for hardware design and component selection while delegating tasks to deliver a high-efficiency Class D amplifier.
- Designed and implemented custom magnetic components using Ansys simulations to meet performance goals.
- Created and assembled multiple PCBs using Altium, improving product quality and testability.
- Troubleshot and debugged custom PCBs under tight design constraints, applying circuit analysis, measurement techniques, and hands-on problem-solving.

Skills

Altium/KiCad	C/C++ & Python	MATLAB/Data Analysis
Linux/GitHub/Scripting	Solid Edge/ Mechanical Design	Signal Processing
Communication & Teamwork	Hardware Troubleshooting	VHDL/Verilog

Education

Utah State University – Logan Utah

- M.S. Electrical Engineering – 3.83 GPA May 2026
- B.S. Electrical Engineering – 3.97 GPA May 2025